MOBILE ARCHITECTURE LAB 4 / MARTECH

Campaign Review & Approval Platform

ONE-PAGE SOLUTION BRIEF | Draft v1.0 | 01 November 2026

THE PROBLEM

Marketing campaign proposals move across marketers, reviewers and approvers before they can be scheduled. When review is coordinated through disconnected tools or informal communication, teams can lose track of the current campaign state, approve one version and act on another, miss required revisions, or lack evidence of who approved what. Concurrent edits and uncontrolled post-approval changes further undermine governance, brand safety, budget control and auditability.

PROPOSED SOLUTION

A mobile-first, workflow-driven platform that manages a campaign proposal from draft through review, revision and approval, while enforcing a scheduling gate. Every review decision is linked to an immutable campaign version; only the currently approved version may be scheduled. The platform provides role-based actions, version history and an append-only record of significant lifecycle events.

CORE USER JOURNEY

1 CREATE Marketer enters objective, audience, budget, schedule and campaign details.	**2 SUBMIT** Validate required fields and create a versioned review snapshot.	**3 REVIEW** Reviewer approves, rejects or requests changes with feedback.
4 REVISE Marketer addresses feedback; a new version is created and resubmitted.	**5 APPROVE** Decision is bound to the exact version reviewed.	**6 SCHEDULE GATE** Scheduling is allowed only when the current version is approved.

ARCHITECTURAL APPROACH

Explicit lifecycle state machine Controlled transitions across Draft, Pending Review, Changes Requested, Approved, Rejected and Scheduled.	**Immutable version snapshots** Review decisions reference a specific campaign version; edits cannot silently change the reviewed proposal.
Consistency & concurrency controls Atomic state transitions, stale-version checks and conflict handling protect approval integrity.	**Reliable side effects** Idempotent operations and asynchronous, retryable notifications prevent delivery failures from undoing business decisions.
Security & separation of duties Authenticated, server-side role checks; submitters cannot review their own campaigns.	**Auditability & observability** Append-only lifecycle history and structured diagnostic logs support reconstruction and troubleshooting.

SCOPE & BOUNDARIES

IN SCOPE Campaign proposal metadata; create/edit/submit; review, approve, reject and request changes; revision and resubmission; version management; scheduling eligibility gate; RBAC; audit history.	**OUT OF SCOPE** Creative/content authoring tools; audience segmentation engine; actual email/ad/push delivery; analytics; A/B testing; integrations with specific CRM, DAM or advertising platforms.

QUALITY GOALS & VALIDATION

Consistency: no stale or contradictory approvals. **Reliability:** retries do not duplicate decisions; notification failure does not roll back approval. **Security:** authentication, server-side authorisation and self-review prevention. **Auditability:** reconstruct every lifecycle action. **Performance:** provisional p95 targets of under 1 second for submit/approve. **Scale assumption:** 5–50 users and 10–200 active campaigns initially.

Validation: exercise the complete mobile core journey, version mismatch at scheduling, concurrent approvals/edits, duplicate retries, rejected and revised submissions, notification failure and audit-log reconstruction.

Working assumptions: initial release uses one authorised reviewer approval; multi-stage/consensus approval is a future extension. Any post-approval edit is treated as material and requires reapproval in the initial design. Multi-tenancy and exact workload targets remain open decisions. Architecture style and technology choices will be justified through HLD and ADRs, not assumed here.
Design principle: An approval belongs to one immutable campaign version; scheduling must verify that exact version. Architecture-led design, supported by a minimal runnable mobile core journey for MAL4.